

Fangyu HOU

Applied Physics Researcher

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Profile

Applied physics researcher working across multimodal olfactory sensing, biophysics, functional materials, scientific computing, and human-centered AI.

Education

Mas- **City University of Hong Kong**, Applied Physics Hong Kong, China
ter of • Dissertation supervisor: Xiangqiang Chu Sept 2023 – Oct 2024
Sci-
ence

Bach- **Lanzhou University**, Physics Lanzhou, China
elor • Specialization: Metallic Materials Physics Sept 2019 – July 2023
of Sci- • Undergraduate thesis supervisor: Dongsheng Lei
ence • Outstanding Student League Cadre, 2020 and 2021 (Top 5%)

Experience

The Future Lab, Tsinghua University, Research Assistant Beijing, China
 Multimodal Olfactory Sensing and Computing Jan 2026 – present
 10 months

- Led a cotton-smoldering early-warning study combining field investigations, AHP-based risk modeling, and TG-FTIR-GC-MS characterization to guide multimodal sensor and experimental-system design.
- Optimized thin-film sensing layers and electrode/substrate architectures using spin coating, layer-by-layer assembly, and ultrasonic spray coating.
- Improved room-temperature NH₃ sensing to reproducible cycles at 10-25 ppm and achieved measurable responses to H₂S and CH₃SH.
- Co-developed E-Nose Station for response-curve visualization, cross-experiment comparison, odor-fingerprint analysis, model-assisted screening, and semi-automated acquisition workflows.

Great Bay University, Research Assistant Dongguan, China
 Biomimetic Colored Radiative-Cooling Films Sept 2025 – Dec 2025
 4 months

- Fabricated and characterized microstructured colored radiative-cooling films using vacuum deposition, UV-Vis spectroscopy, infrared and thermal imaging, nanoindentation, and microscopy.
- Processed spectral and imaging data to analyze relationships among microstructure, visible appearance, optical transparency, and thermal behavior.

Institute of Physics, Chinese Academy of Sciences, Research Assistant Beijing, China
 GSDM Protein Dynamics and Cell Pyroptosis Sept 2024 – Dec 2024
 4 months

- Used HS-AFM to visualize pore-forming dynamics of ThGSDM, GSDMD, and GSDME on bare mica and DOPC/DOPS supported lipid bilayers.
- Contributed to X-ray crystallography data processing.

National Undergraduate Innovation and Entrepreneurship Training Program, Undergraduate Researcher
Flexible-Substrate Metasurface Optical Tweezers

- Designed PDMS flexible-substrate metasurface lenses and nanopillar unit cells using FDTD simulations and MATLAB optical-force modeling.
- Modeled substrate deformation and demonstrated stable simulated trapping of 15 nm polystyrene spheres near the self-focusing region.

Lanzhou, China
Mar 2022 – Mar 2023
1 year 1 month

Projects

Locus: Human-AI Decision Support for PhD Research Fit and Advisor Discovery

Independent project

- Designed and developed a human-AI decision-support platform for personalized PhD research positioning and advisor exploration.
- Built an adaptive advisor search and ranking engine using offline faculty retrieval, LLM-generated candidates, research-similarity scoring, and institution and discipline constraints.
- Added uncertainty-aware verification using web, OpenAlex, and ORCID evidence and piloted the platform with more than 50 users.

Beijing, China
Apr 2026 – present

Publications

Computational Design of 3D-Printable Freeform Caustic Surfaces

Shuowen Li, Danqi Huang, *Fangyu Hou*, Qiuyu Lu, Haipeng Mi, Zhihao Yao

[10.1145/3799825.3818720](https://doi.org/10.1145/3799825.3818720) (ACM SIGGRAPH 2026 Posters, Article 4)

Jan 2026

Patents

1. Cotton Smoldering Simulation and Multimodal Gas Detection Device - Chinese invention patent application, pending (2026)

Skills

Programming and Data Analysis: Python (NumPy, Pandas, Matplotlib, SciPy), MATLAB, SQLite

Web Development and Deployment: JavaScript, HTML/CSS, Flask, Linux, Git, Nginx, web deployment, DNS, SSL/TLS configuration

Scientific Computing and Engineering: FDTD Solutions, COMSOL, SolidWorks, Arduino

Sensing and Experimental Methods: HS-AFM/AFM, spin coating, ultrasonic spray coating, layer-by-layer assembly, and optical, thermal, and mechanical materials characterization